SCORING GUIDE 2026

Candidate's Name..... MR. LAP .....

Signature..... .....Index number.....

P525/2

CHEMISTRY

Paper 2

JULY/AUGUST-2026

3 ¼ HOURS

JINJA JOINT EXAMINATIONS BOARD

UGANDA ADVANCED CERTIFICATE OF EDUCATION
CHEMISTRY PRACTICAL
3 HOURS 15 MINUTES

INSTRUCTIONS TO CANDIDATES:

- This paper consists of two compulsory examination items. Responses to these items are to be written in the spaces provided in this booklet.
- Use blue or black ink.
- All working must be clearly shown. Graph paper will be provided.
- Mathematical table and silent non-programmable scientific calculators may be used.
- You are not allowed to used reference books (i.e. text books, booklets on qualitative analysis etc.)
- Candidates are advised to carefully read the items, make sure they have all the apparatus and chemicals they may need and then plan appropriately before starting.
- QUESTIONS ATTEMPTED

Question Number	For examiner's use only	
	Mark	Examiner's initial
ITEM 1	PP/PE = 03	LAP
	ME = 03	LAP
	DP1 = 03	LAP
	AN1 = 03	LAP
	CR = 02	LAP
	SCORE = 14	LAP
ITEM 2	QP2 = 03	LAP
	QO2 = 03	LAP
	QD2 = 03	LAP
	CR2 = 02	LAP
	SCORE = 11	LAP
TOTAL SCORE	25	LAP

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ITEM 1

Iron supplements are used in treatment of patients who are anemic especially pregnant mothers and malnourished children. One of the key ingredients a Pharmaceutical Company in Luzira uses in the production of the supplements is hydrated iron (II) salt of the form $FeSO_4.nH_2O$. The Company claims that the specific water molecules n, attached to each iron (II) salt unit should be in range of 5-8 for it to be effective.

A certain batch of the ingredient has overstayed in the stores of the Company for a long period of time and they are fears that the salt may have lost the water of Crystallization, making it ineffective or unsuitable in the production of the iron Supplement.

In order to determine the exact value of n in the hydrated Salt, potassium Manganate(VII) Solution is required to quantitatively react with a solution of acidified Iron(II) ions from the hydrated Iron(II) salt.

However the potassium Manganate Solution available in the laboratory has been there for some time and its Concentration may have deteriorated. Therefore the potassium Manganate Solution must first be standardized using a ***hot*** *solution of acidified oxalic acid, made by dissolving* ***exactly*** *1.6g of $H_2C_2O_4.2H_2O$ in distilled water and making the Solution up to 250cm³ before it is used. The Company experiences some challenges in carrying out the analysis and has sought assistance from a nearby school laboratory.*

Manganate(VII) ions in solution react with Iron(II) ions and oxalate ions in acidic medium according to the following equations:

$$`MnO_4^-(aq) + 5Fe^{2+}(aq) + 8H^+(aq) \rightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O`$$

$$`2MnO_4^-(aq) + 5C_2O_4^{2-}(aq) + 16H^+(aq) \rightarrow 2Mn^{2+}(aq) + 8H_2O + 10CO_2`$$

You are provided with the following;

FA1; which is a sample Solution of a batch of the ingredient made by dissolving 8.8g of hydrated (II) Salt, $FeSO_4.nH_2O$ in 250cm³ of water.

FA2; which is potassium Manganate (VII) Solution.

FA3; 2M Sulphuric acid

Solid P, which is oxalic acid, $H_2C_2O_4.2H_2O$

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distilled water

Thermometer

(C=12, O=16, Fe=56, S=32, H=1);

TASK: *As a chemistry learner,*

(a) Design an experiment to determine the value of n in the batch of the ingredient (your design should include an experimental plan, procedure, risks and mitigations).

Experimental Planning

Aim: To determine the number of moles of water of crystallization, in a batch of hydrated iron(II) salt intended to find out its suitability or effectiveness as an iron supplement by measuring the volume of potassium permanganate that reacts with a fixed volume of iron(II) ions in a solution prepared by dissolving 8.8g of the hydrated salt in 250 cm^3.

Hypothesis: The exact value of the number of moles of crystallization, n, is in the range of 5-8, determined by measuring the volume of potassium permanganate that reacts with a fixed volume of iron(II) ions pipetted.

Variables: Dependent: Volume of potassium permanganate solution which completely reacts with iron(II) ions.
Independent: Volume of iron(II) ions solution.
Controlled: Volume of iron(II) solution pipetted, kept constant at 10 cm^3
8.8g of hydrated iron(II) salt.

E.P
Score: 3V + 4A + 3H = 03
3V + 4A = 02
3V = 01
Reject scores for <3V

Method of Experiment
Procedure
1) A 50 cm^3 burette was rinsed with some FA2 then clamped with a retort stand in an upright position, filled with FA2 while ensuring air bubbles was tapped in the tap. The initial volume was taken and recorded. Pr
2) 1.6g of P was accurately weighed using a weighing electrical balance (scale) Pr and placed in a beaker containing about 50 cm^3 of distilled water.

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stirred using a stirring rod. Pr

3) The solution in (2) was quantitatively transferred into a 250 cm³ volumetric flask and more distilled water was added ~~into a 250 cm³~~ to make up to the make up to the mark. Pr The resultant solution was labelled FA4.

4) A 10 cm³ pipette was rinsed with some FA4 and 10 cm³ of FA4 was transferred into a clean conical flask. Pr

5) Using a measuring cyclinder, 10 cm³ of 2M sulphuric acid (FA3) was added to the conical flask in (4) and the mixture heated to a temperature of 60°c to 80°c using a thermometer. Pr

6) Solution FA2 was then run into the hot mixture in the conical flask in (5) while stirring at each interval and the titration continued until a permanent pink coloration persisted on shaking. The final volume was taken and the reading was recorded. By substraction, the volume of FA2 transferred was obtained.

7) Procedure (4) to (6) was repeated two more times until constent results were obtained which were then read, recorded and entered in a table was obtained.

Reject scores for no repetition of procdure 7 (Not working procedure)

8) Using a 10 cm³ pipette, 10 cm³ of FA1 was transferred into a clean conical flask in (8)

9) Using a measuring cyclinder, 10 cm³ of 2M sulphuric acid (FA3) was added to the conical flask in (8)

10) The burette was filled with more FA2 and the initial volume was read and recorded.

11) Solution FA2 was then run in the conical flask in (9), while stirring gently at intervals and the titration continued until a permanent pink coloration persisted on shaking. The final Volume was taken and the reading was recorded. By subtraction, the volume of FA2 transferred was obtained.

12) Procedure (8) to (11) was repeated two more times until consistent results were obtained which were read, recorded and entered in a table of results.

Reject Scores for procedure for no repetition in procedure 12 (Not working procedure)

Score: (13-21) Pr + 2Rm = 03

ME (10-12) Pr = 02

(7-9) Pr = 01

Mitigated Risks:

Ingestion of iron(II) ions in solution during pipetting cause stomach disorder such as vomiting. Mitigated by use Rm a pipette filler to prevent swallowing of the solution of iron(II) ions

Sulphuric acid spills on skin which is highly corrosive causing severe burns, mitigated by Rm wearing Laboratory coats or gloves to avoid direct contact of acid with skin.

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Pr = relevant apparatus, specify quantity, coherence

Rm = risks mitigated.

(b) Carry out your experiment and record your results.

Mass of empty container + solid P = 2.6 g
Mass of empty container alone = 1.00 g ✓Dr
Mass of P alone = 1.6 g
Volume of pipette used = 10.0 cm³

Table of Standardizing of potassium per manganate, FA_2

Titration Number	1	2	3
Final burette reading (cm³)	10.50	20.70	30.90
Initial burette reading (cm³)	0.00	10.50	20.70 ✓Dr
Volume of FA_2 used (cm³)	10.50	10.20	10.20 ✓Ds ✓Dc ✓Dc

Table for Determining the value of crystallization

Titration Number	1	2	3
Final burette reading (cm³)	14.00	26.70	39.40 ✓Dr ✓Ds
Initial burette reading (cm³)	1.00	14.00	26.70 ✓Ds ✓Dc
Volume of FA_2 used (cm³)	13.00	12.70	12.70 ✓Dc

Dr = Variables with units in a table
Dc = Accuracy and number of decimal points
Ds = 3 data sets

Score; 3Dr + 2Ds + 4Dc = 03
(DP1) 2Dr + 2Ds + 2Dc = 02
2Dr + 2Ds = 01

(c) Analyse your results appropriately.

Standardizing of potassium per manganate in FA_2
Average volume of FA_2 = $\frac{10.2 + 10.2}{2}$ = 10.2 ✓Da

RFM of $H_2C_2O_4.2H_2O$ = 2 + 2(12) + 4(16) + 2(18)
= 2 + 24 + 64 + 36 ✓Da
= 126

126 g of $H_2C_2O_4.2H_2O$ contain 1 mole of ✓Dc
1.6 g of $H_2C_2O_4.2H_2O$ contain $\left(1 \times \frac{1.6}{126}\right)$ of $H_2C_2O_4.2H_2O$
= 0.0127 moles of $C_2O_4^{2-}$ $H_2C_2O_4.2H_2O$

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250cm³ of FA_4 contain 0.0127 moles of $C_2O_4^{2-}$

10 cm³ of FA_4 contain $\left(\frac{0.0127 \times 10}{250}\right)$ moles of $C_2O_4^{2-}$ Da

$= 5.08 \times 10^{-4}$ moles of $C_2O_4^{2-}$

From equation:

5 moles of $C_2O_4^{2-}$ react with 2 moles of MnO_4^- Da

5.08×10^{-4} moles of $C_2O_4^{2-}$ react with $\left(\frac{2 \times 5.08 \times 10^{-4}}{5}\right)$ moles of MnO_4^-

$= 2.032 \times 10^{-4}$ moles of MnO_4^-

10.2 cm³ of FA_2 contain 2.032×10^{-4} moles of MnO_4^-

1000 cm³ of FA_2 contain $\left(\frac{2.032 \times 10^{-4} \times 1000}{10.2}\right)$ moles of MnO_4^- Da

$= 0.0199$ M of MnO_4^- Da

Determining of Value of n in $FeSO_4 \cdot nH_2O$

Average volume of $FA_2 = \left(\frac{12.7 + 12.7}{2}\right)$ Da $= 12.7$

1000cm³ of FA_2 contain 0.0199 moles of MnO_4^-

12.7 cm³ of FA2 contain $\left(\frac{0.0199 \times 12.7}{1000}\right)$ moles of MnO_4^- Da

$= 2.527 \times 10^{-4}$ moles of MnO_4^-

1 mole of MnO_4^- reacts with 5 moles of Fe^{2+} Da

2.52×10^{-4} moles of MnO_4^- react with $(5 \times 2.52 \times 10^{-4})$ of Fe^{2+}

$= 1.264 \times 10^{-3}$ moles of Fe^{2+}

10 cm³ of FA_4 contain 1.264×10^{-3} moles of Fe^{2+}

250 cm³ of FA_4 contain $\left(\frac{1.26 \times 10^{-3} \times 250}{10}\right)$ moles of Fe^{2+} Da

$= 0.0316$ moles of Fe^{2+}

0.0316 Moles of Fe^{2+} contain 8.8 g

1 mole of Fe^{2+} contain $\left(\frac{1 \times 8.8}{0.0316}\right)$ Da

$= 278.48$ Da

$FeSO_4 \cdot nH_2O = 278.48$ Da

$56 + 32 + 64 + 18n = 278.48$

$18n = 278.48 - 152 = 126.48$

$\frac{18n}{18} = \frac{126.48}{18}$ Da

$n = 7$ Da

Value of n = 7 which is within the claimed range for n of 5 to 8: A1

$(11 - 16)Da + Di = 02$

$(11 - 16)Da = 02$

$5 - 7\,Da = 01$

(d) What conclusion and recommendations do you make from your findings?

Conclusion:

The exact value of number of moles of water of crystallization, n is 7 which is within the manufacturer's claim

Recommendations

The Batch of ingredients in the store can be used in the production of iron supplements for the treatment of anaemic people

CR1 = C + R = 02

C = 01

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C = Conclusion

R = recommendation

ME = 03 Scores

PE = 03 Scores

DP1 = 03 Scores

AN1 = 03 Scores

CR1 = 02 Scores

TOTAL SCORE = 14 Scores

ITEM 2

Ionic salts have a wide range of uses in daily life across many fields, such as manufacturing and agriculture. However, they must be handled with extra caution because they can cause significant health and environmental challenges, including water and soil pollution.

A small stream behind an industrial estate in Soroti District has turned cloudy white. Local residents have reported that the water feels "slimy" and has caused white, powdery deposits on the stones near the discharge pipe. The District Health Office suspects that an unauthorized workshop—either a local acid battery refurbishing site or an illegal automobile paint-spraying operation—is dumping its liquid waste directly into the community drainage system.

The community intends to revert to using the water for small-scale irrigation and livestock, as other available water sources have dried up due to the prolonged drought. Therefore, the community wants to determine whether it is advisable to use this water source.

*You have been provided with **Sample X**, a mixture of solid sludge scraped from a discharge point. Your objective is to perform a systematic qualitative analysis to identify the presence of specific ions in **X** and, consequently, the salts likely to have polluted the water.*

***Sample X** is sparingly soluble in water and contains **two cations** and **two anions**.*

Task

***(a)** Plan an experimental investigation to analyze the sample. (Your plan should include a **minimum of eleven tests**.)*

***(b)** Record your observations and make corresponding deductions.*

***(c)** State your conclusion and, based on your findings, provide recommendation(s) on whether this water is safe for irrigation or livestock.*

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Tests	Observations	Deductions
a) One spatula endful of X was heated first gently and then strongly until no further change	White solid decomposed to colourless condensate that turns white anhydrous copper(II) sulphate blue. Colourless gas turned wet blue litmus red and lime water milky. Residue was reddish brown (orange when hot and yellow when cold)	Probably Ba^{2+} or Pb^{2+} or Ca^{2+} or Mg^{2+} present. Hydrated salt. Gas was CO_2. CO_3^{2-} or HCO_3^- or $C_2O_4^{2-}$ present (Any one). PbO: Pb^{2+}
b) To one spatula endful of X was added 4 cm³ of water shaken vigorously and filtered. The residue was kept for part (c) while the filtrate was divided into 6 parts.	Partly soluble. White residue. Colourless filtrate was formed.	Probably Non transition metal ions in both filtrate and residue.
i) To the first part of the filtrate was added sodium hydroxide solution dropwise until in excess	White precipitate insoluble in excess was formed.	Probably Mg^{2+}, Ca^{2+}, Ba^{2+} present. Any 2. For one wrong cancels the correct one.
ii) To the second part of the filtrate was added ammonia solution dropwise until in excess	White precipitate insoluble in excess was formed.	Probably Ba^{2+} or Mg^{2+} present.
iii) To the third part of the filtrate was added 3 drops of dilute sulphuric acid	White precipitate was formed.	Probably Ba^{2+} present
iv) To the fourth part of the filtrate was added 4 drops of potassium chromate solution followed by sodium hydroxide dropwise until in excess	Yellow precipitate insoluble in excess was formed.	Ba^{2+} confirmed.

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v) To the fifth part of the filtrate was added 3 drops of nitric acid followed by 5 drops of lead(II) nitrate solution and warmed. P_5 Reject without the acid	White precipitate soluble on warming O_5	Cl^- present D_5
vi) To the sixth part of the filtrate was added 3 drops of nitric acid followed by 5 drops of silver nitrate solution. P_6	White precipitate was formed O_6	Cl^- confirmed D_6
c) The residue from part b was washed with some water and then dissolved in about 5 cm^3 of dilute nitric acid. P_7 The resultant mixture was divided into 4 portions Reject: dil HCl / H_2SO_4	Effervescence or bubbles of colourless gas which turn wet blue litmus red and lime water milky O_7 Colourless solution formed O_8	Gas is CO_2 (g) CO_3^{2-} D_7 Probably non transition metal ions present D_8
i) To the first portion of the solution was added dilute sodium hydroxide solution dropwise until in excess P_8	White precipitate soluble in excess forming a colorless solution O_9	Probably Zn^{2+} or Al^{3+} or Pb^{2+} present D_9 D_{10}
ii) To the second portion was added ammonia solution dropwise until in excess P_9	White precipitate insoluble in excess O_{10}	Probably Al^{3+} or Pb^{2+} present D_{11} D_{12}
iii) To the third portion was added 4 drops of dilute sulphuric acid	White precipitate was formed O_{11}	Probably Pb^{2+} present D_{13}
iv) To the fourth part was added 5 drops of potassium chromate solution followed by dilute sodium hydroxide solution dropwise until in excess P_{10} OR Added 4 drops of potassium iodide solution	Yellow precipitate soluble in excess sodium hydroxide solution O_{12} Yellow ppt	Pb^{2+} Confirmed D_{14}

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Conclusion:

Cations in X are Ba^{2+} and Pb^{2+}

Anions in X are Cl^- and CO_3^{2-}

OR

Sample X is a mixture of $PbCO_3$ and $BaCl_2$

Recommendations

The metals are toxic, which accumulate in the soil, plants, animals and humans causing diseases and death. The farmers should find an alternative source of water such as rain water harvesting or treats the water.

Scoring

Planning Tests: (QP2)	$3P_p + 8P_s + 3P_c$	= 03	Scores
	$3P_p + 8P_s$	= 02	Scores
	$3P_p$	= 01	Scores
Making Observations from test (QO2)	$10O_p + 7O_s + 4O_c$	= 03	Scores
	$10O_p + 7O_s$	= 02	Scores
	$10O_p$	= 01	Scores
Making logical Deductions (QD2)	$8D_p + 11D_s + 4D_c$	= 03	Scores
	$8D_p + 11D_s$	= 02	Score
	$8D_p$	= 01	Score

P_p = Preliminary test, P_s = Preliminary specific test, P_c = Confirmatory test.

O_p = Preliminary observations, O_s = observations from specific test

O_c = Observations from confirmatory test

D_p = Deduction from preliminary test

D_s = Deduction from specific test

D_c = Deduction from confirmatory test.

Conclusion and Recommendation (CR2)	C + R	= 02 Score
	C	= 01 Score

TOTAL SCORE = 11 Scores

END

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